Grokking Under a Fixed Budget

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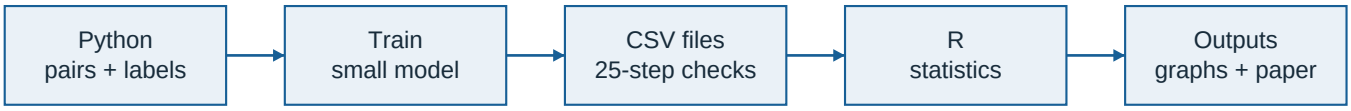
1. The task and dataset

Modulo keeps the remainder

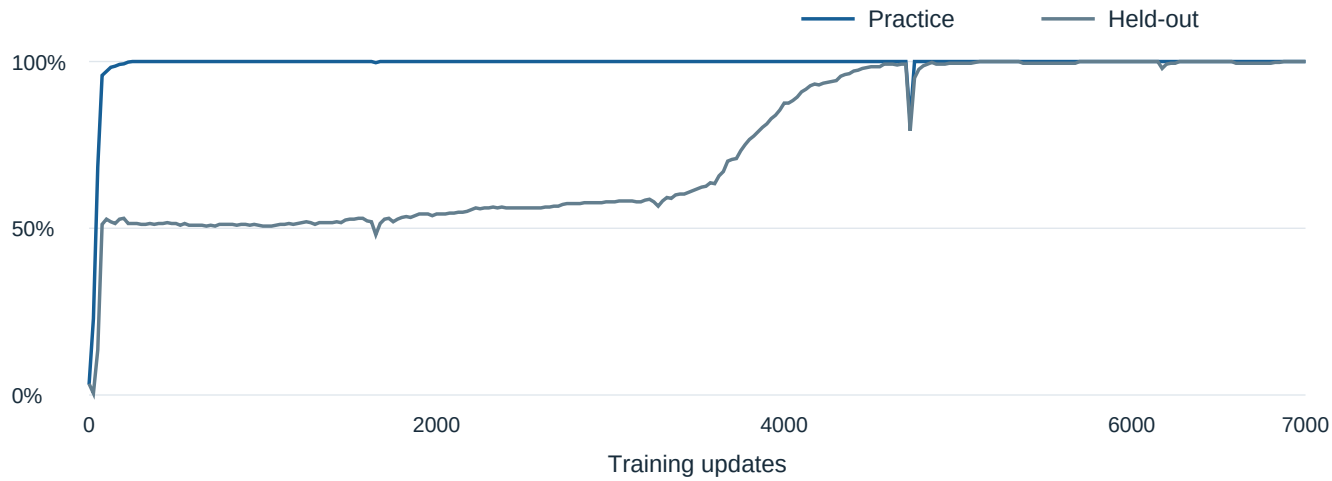
961 pairs. Three fixed splits. Independent starting seeds.

2. How the code works



Checkpoint saves: weights, optimizer and random state.

3. Grokking: practice improves before new-question accuracy



S1 calibration (n=30): mean 5,347; median 4,375; SD 4,610 updates.

4. Two success rules

Every check: practice >=99% and held-out >=95%.

First: 3 checks ● ● ● 50-update span

Window: 21 checks ● 500-update span

Spans start at the first good check. Count the last check inside B/k.

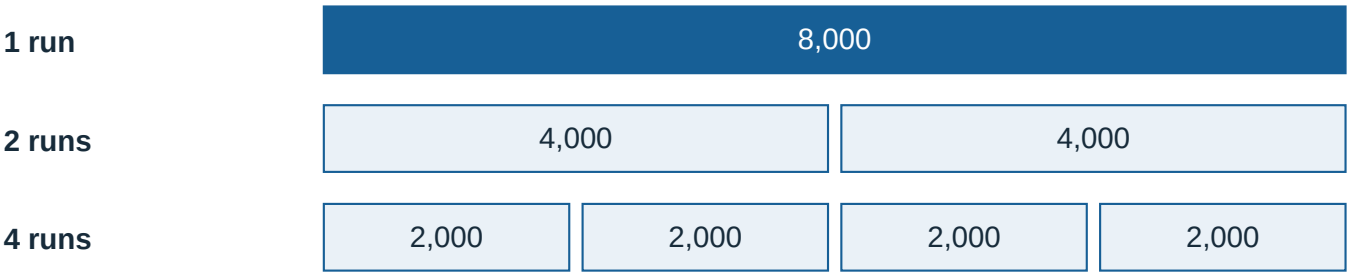
U=min(T,C): capped time; eventual unfinished T stays unknown.

Chance of >=1 success: 1 - [1 - F(B/k)]^k F(t): one-run completion chance.

Results and the decision

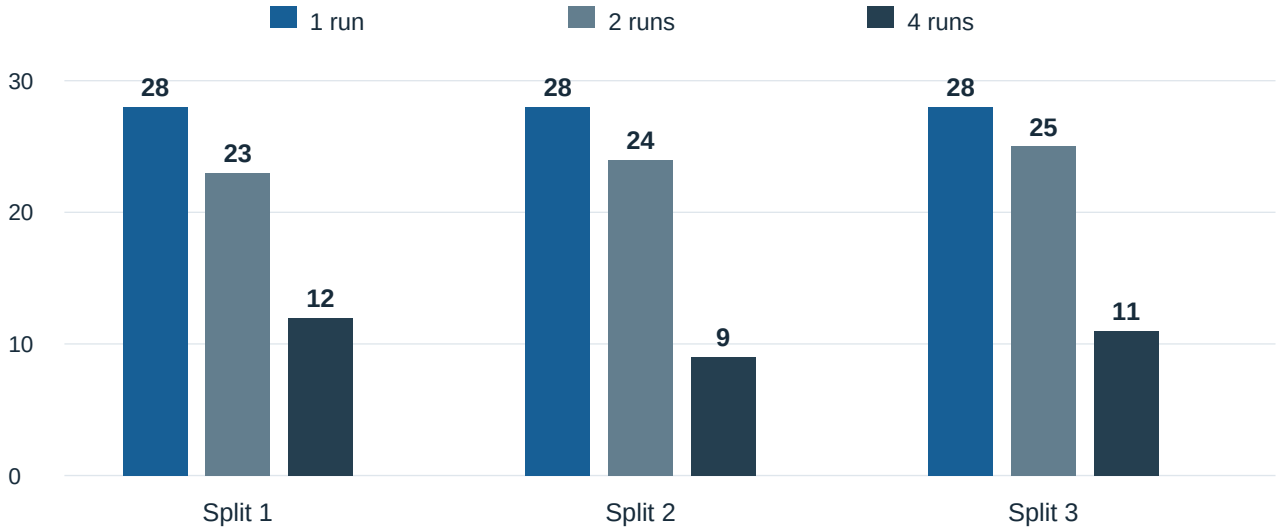
Binary attempts. Exact intervals. Twelve Fisher tests with Holm correction.

5. Same total budget: B = 8,000 updates



Each segment is a fresh start. All allocated updates execute.

6. Fresh successes at the 8,000-update budget



Each bar: successes out of 30 independent attempts.

All three 4-versus-1 differences survive Holm correction.

7. Results at 12,000 updates (out of 30)

Split	1 run	2 runs	4 runs
S1	30	30	24
S2	30	30	20
S3	30	29	23

12k: only S2 detects a 4-vs-1 difference. Two-run results are inconclusive.

Example uncertainty: 28/30 = 93.3%; exact 95% interval 77.9%-99.2%.

8. Brief confirmation versus a longer window

S1, 8k, four starts

Decision: one long run was more reliable at 8k.

Scope: one task, one model, three splits. Later accuracy can still fall.

Next: other tasks, more splits and longer accuracy windows.

Sources: Power et al. (2022); Streeter et al. (2007). Full references: Assessment 9.